CHAPTER

17

New Technologies and Automation

We live in an age of rapid discovery in biotechnology, and new technologies and instruments are continually being developed. Any attempt to predict in detail where the field of forensic DNA typing will be in five or ten years would be futile given the rate of technology change. Because of parallel efforts in biotechnology development (e.g., genome sequencing), there is growing set of tools for forensic DNA analysts to use, many of which are described in this book.

The forensic DNA field has advanced quickly over the past several decades but has now stabilized on short tandem repeat (STR) typing with capillary electrophoresis detection. However, a comparison of the new information found in this book with previous editions of *Forensic DNA Typing* illustrates that innovations are being made with almost every step along the process of producing a DNA profile. The goal of this chapter is to briefly discuss some technologies with potential impact on future forensic DNA analysis and the value of automation to increase sample throughput.

Similar to the Olympic motto of "faster, higher, stronger," there is a desire for faster DNA analysis, higher sensitivity in detection, and stronger powers of discrimination especially with related individuals. Rapid DNA analysis efforts, improved allelic resolution with mass spectrometry, and the capabilities and challenges of next-generation sequencing technology are discussed below as are some ongoing efforts with automation involving robotic liquid handling, laboratory information management systems, and expert systems for data review.

RAPID DNA ANALYSIS

The DNA testing process outlined in Chapter 1 to Chapter 6 can be performed in as little as 8 to 10 hours with the longest part of the process being the 3-hour PCR amplification step. As mentioned in Chapter 4, rapid PCR efforts using specialized DNA polymerases and faster temperature ramp rates have brought multiplex PCR amplification times down to around 30 minutes or less (Vallone et al. 2008, Giese et al. 2009).

Advanced Topics in Forensic DNA Typing: Methodology.
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TABLE 17.1 Groups Involved in Rapid DNA Analysis Efforts. For Additional Information (Provided as of Sept 2010), See Presentations from Special Rapid DNA Session at http://www.biometrics.org/bc2010/program.pdf.

Company or Group	Website/Reference
Forensic Science Service/ University of Arizona	Hopwood et al. (2010)
IntegenX	www.integenx.com
Lockheed Martin/ZyGEM/ MicroLab Diagnostics	www.lockheedmartin.com www.zygem.com Bienvenue et al. (2010)
Network Biosystems	www.netbio.com Giese et al. (2009)

Efforts are underway to integrate the various steps involved in DNA extraction, amplification, and STR allele detection. The primary players (as of early 2011) in the effort to develop rapid DNA analysis devices are listed in Table 17.1. Routine STR typing in less than an hour is a strong possiblity in the near future. Rapid analysis of DNA could open up a whole new set of DNA biometric applications such as analysis of individuals at a point of interest like an airport or a country border. Only time will tell to what extent technology will enable new applications.

The desire to take DNA testing capabilities out of the laboratory to a crime scene or close to a battlefield is propelling efforts into development of portable DNA testing devices. For example, some have claimed that a rapid DNA device might aid elimination of innocent suspects early in an investigation. Much of the work so far has been focused on miniaturizing the DNA separation steps.

Microfabrication techniques revolutionized the integrated circuit industry 20 years ago and have brought the world ever-faster and more powerful computers. These same microfabrication methods are now being applied to develop miniature, microchip-based laboratories, or so-called "labs-on-a-chip" (Paegel et al. 2003). Miniaturizing the sample preparation and analysis steps in forensic DNA typing could lead to devices that permit investigation of biological evidence at a crime scene or more rapid and less expensive DNA analysis in a conventional laboratory setting.

The primary advantage of analyzing DNA in a miniature capillary electrophoresis (CE) device is that shorter channels, or capillaries, lead to faster DNA separations. Separation speeds that are 10 to 100 times faster than conventional electrophoresis may be obtained with this approach (Figure 17.1). Over a decade ago, tetranucleotide STR alleles were separated in as little as 30 seconds (Schmalzing et al. 1997) using a 2-cm separation distance (compared to 36-cm for an ABI 310 capillary). However, routine and robust analyses of DNA with appropriate resolution of STR alleles have not yet been achieved at these speeds even though several groups are working extensively in this area.

Research is ongoing to improve separation speeds and ease of use with the hope that in the near future microchip CE devices will be used routinely for rapid DNA analyses. A report of DNA analysis performed in a mobile van in less than six hours at a mock crime scene was published in September 2008 (Liu et al. 2008). When portable devices become

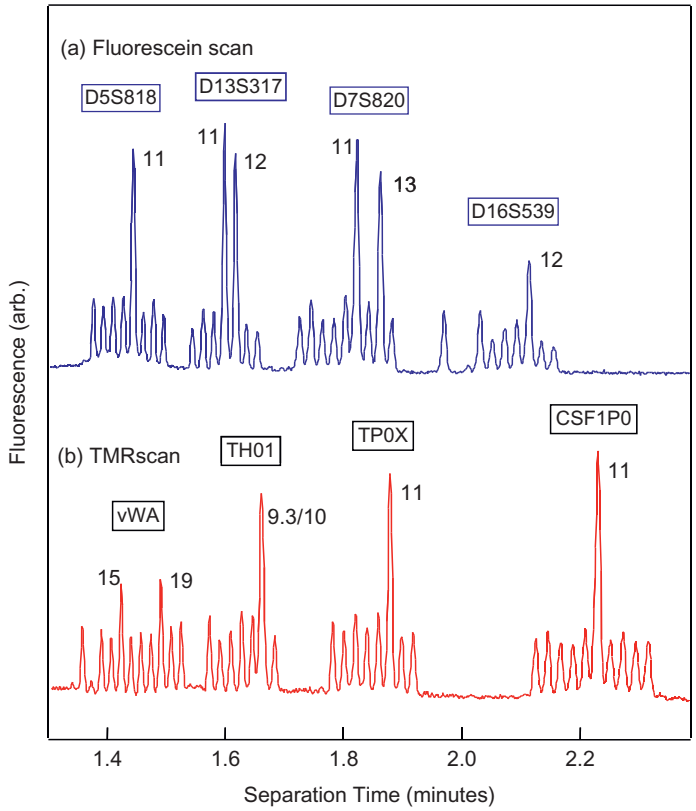


FIGURE 17.1 Rapid microchip CE separation of the 8 STR loci from PowerPlex 1.1 (Schmalzing et al. 1999). The electropherograms from scanning each color of a simultaneous two-color analysis are divided by color. The PCR-amplified sample is mixed with the allelic ladders prior to injection to provide a frame of reference for genotyping the sample. The allele calls for each locus are listed next to the corresponding peak. Figure courtesy of Dr. Daniel Ehrlich, Whitehead Institute.

available, there will be a need for sensitivity and resolution standards to make sure that DNA separation and detection devices at different locations have performance similar to one another.

A major challenge is the integration of DNA processing steps so that a "swab-to-profile" result can be obtained without any user intervention. The Forensic Science Service and the University of Arizona Center for Applied NanoBioscience and Medicine reported such an integrated device in August 2010 (Hopwood et al. 2010). Results were achieved in less than four hours with a DNA processing cartridge with wax seals that delivered sample and reagents to the necessary reaction chambers to permit DNA purification, PCR amplification, and collection of the amplified product. An accompanying CE chip connected by Telfon tubing performed the DNA separation (Hopwood et al. 2010, Hurth et al. 2010).

Richard Mathies' group at the University of California-Berkeley has been working on integration of PCR and microchip CE systems for years (Woolley et al. 1996) and has

demonstrated a three hour integrated swab-to-profile result (Liu et al. 2011). Lockheed Martin and ZyGEM Microlab are also showing promising results towards full sample process integration with their RapI.D. system (Bienvenue et al. 2010, Reedy et al. 2011).

MASS SPECTROMETRY

Mass spectrometry is a versatile analytical technique that involves the detection of ions and the measurement of their mass-to-charge ratio. Because these ions are separated in a vacuum environment, the analysis times can be extremely rapid, on the order of seconds. Combined with robotic sample preparation, mass spectrometry offers the potential for processing vast numbers of DNA samples in an automated fashion.

With mass spectrometry, the actual mass of the DNA molecule is being measured, making it a more accurate technique than a relative-size measurement as is obtained in electrophoresis. In order to get the DNA molecules into the gas phase for analysis in the mass spectrometer, two different ionization techniques have been used: matrix-assisted laser desorption-ionization (MALDI) or electrospray ionization (ESI).

Over a decade ago, STR markers were successfully analyzed via MALDI time-of-flight mass spectrometry by redesigning the PCR primers to bind closer to the repeat region, and thereby reducing the size of the amplified alleles (Butler et al. 1998). While MALDI had a size limitation of ≈100 bp, recent ESI time-of-flight techniques have extended the size range for accurate mass spectrometry measurements to ≈250 bp.

Two ESI mass spectrometry approaches are in use: (1) ion-pair reversed-phase liquid chromatography coupled with an ESI quadrupole time-of-flight (qTOF) mass analyzer

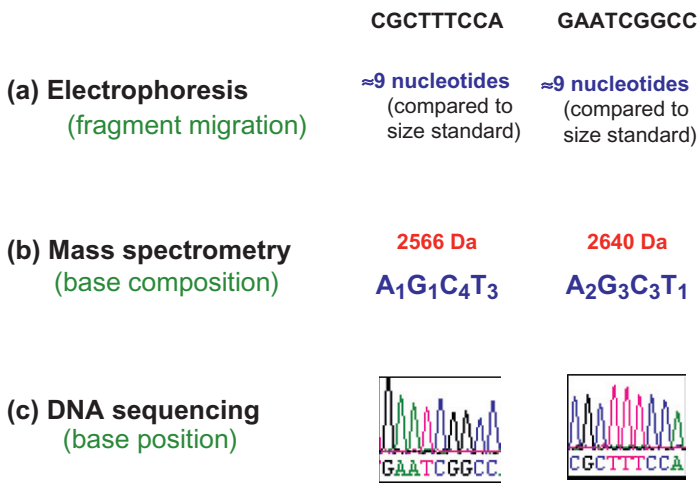


FIGURE 17.2 Comparison of measurement techniques with the ability to resolve two 9-base sequences. (a) Electrophoresis measures DNA fragment migration relative to a size standard. (b) Mass spectrometry measures the DNA fragment mass and can determine base composition using mass measurement conversion to numbers of individual nucleotides, which have different individual masses. (c) DNA sequencing measures the exact position of bases within a DNA fragment.

(Oberacher & Parson 2007) and (2) sample desalting prior to an ESI-Fourier transform ion cyclotron resonance (FT-ICR) (Hall et al. 2005, Jiang et al. 2007) or an ESI-time-of-flight (Hall et al. 2009) mass analyzer.

By analyzing the overall mass of an STR allele or a section of mtDNA, base composition of the measured DNA molecule can be deciphered and internal sequence differences ascertained (Hall et al. 2005, Oberacher & Parson 2007). STR alleles that are apparently homozygous by electrophoretic techiques have been subdivided into separate/unique alleles when internal sequence polymorphisms exist (Oberacher et al. 2008). However, the exact position of a sequence difference cannot be assessed with mass spectrometry (Figure 17.2).

The expense of mass spectrometers, the expertise required to keep them running, and the previous wide-scale acceptance of electrophoretic separation and fluorescence detection methodologies will likely keep mass spectrometry from becoming a major player in forensic DNA analysis of STR markers. However, capabilities for mitochondrial DNA base composition analysis and single nucleotide polymorphism (SNP) typing may enable mass spectrometry to play a useful role in the future of forensic DNA analysis.

PYROSEQUENCING

Pyrosequencing is based on detection of pyrophosphate released when a deoxynucleotide triphosphate (dNTP) is incorporated into a growing DNA strand (Ronaghi et al. 1998). In this sequencing-by-synthesis approach, dNTPs are added in a stepwise fashion through a directed dispensation (i.e., the user controls which nucleotide—A, T, C, or G—is added). If the nucleotide is incorporated (due to being a match to the complementary target DNA strand), pyrophosphate is released. Through an enzymatic reaction cascade involving the formation of ATP and luciferin being converted to oxyluciferin, light is emitted. Excess dNTP is degraded by apyrase prior to the addition of the next nucleotide. The light emitted is detected in a pyrogram where peak heights are proportional to the number of incorporated nucleotides (Figure 17.3).

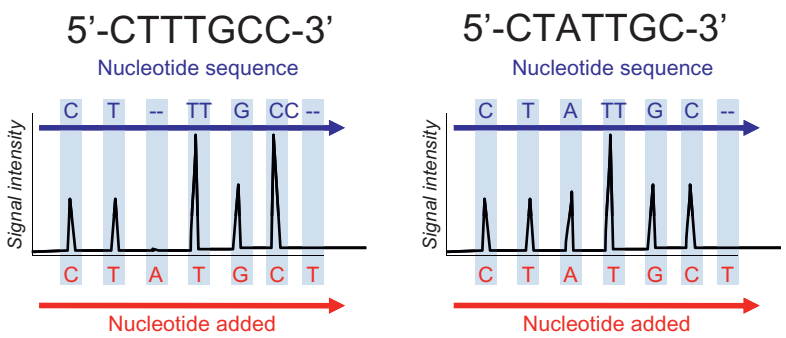


FIGURE 17.3 Schematic illustration of Pyrosequencing (sequencing-by-synthesis) results. These pyrograms are based on light produced and detected during nucleotide incorporation following a directed dispensation of nucleotides.

Pyrosequencing has been used for mitochondrial DNA sequence analysis (Andreasson et al. 2002, Allen & Andreason 2005), Y-STR typing (Edlund & Allen 2009), amelogenin sex typing (Tschentscher et al. 2008), autosomal STR typing (Divne et al. 2010), and SNP analysis (Zeng et al. 2010).

NEXT-GENERATION DNA SEQUENCING

Next-generation DNA sequencing involves rapid, high-throughput collection of short sections of DNA. This approach was first introduced in 2003 by the 454 Company. The desire for more genomic information is improving technology and reducing costs. A motivating goal for this new technology is to obtain a full human genome sequence for $1000 or less. Because these next-generation sequencing approaches generate short sequence reads of typically 25 to 250 bases in a massively parallel fashion, it has been possible to successfully obtain sequence information from ancient DNA samples, which are normally too fragmented to sequence by traditional techniques. A complete Neandertal mitochondrial genome sequence has been obtained (Green et al. 2008) as has DNA sequence information from the extinct woolly mammoth (Miller et al. 2008, Poinar et al. 2006).

It is unclear whether or not next-generation sequencing techniques will help traditional forensic testing as they are helping ancient DNA sample sequencing (Blow et al. 2008). Current methods have a difficult time with repetitive sequences and thus unless future improvements are made, STR regions would probably not be reliably analyzed with next-generation DNA sequencing (Hert et al. 2008). In addition, the amount of data produced by these next-generation systems with millions of short reads makes bioinformatics support crucial and would require a significant change in the way that forensic DNA laboratories do business as well as a switch of genetic markers from STRs to SNPs.

LABORATORY AUTOMATION

Laboratory automation is an important topic, especially because the demand for forensic DNA testing is increasing. Laboratories will take on more cases and have much larger numbers of samples to type because of expanding DNA database laws and more extensive use of DNA in casework applications. Automation aids efficiency and error reduction. Often sample throughput can be increased and total processing time can be reduced when a more consistent process is introduced through automation. Reducing analyst involvement in routine sample processing can enable more focus on evidence examination and data interpretation. In addition, automation reduces potential errors through better sample tracking and handling.

The use of laboratory automation in DNA typing laboratories is growing in several areas: liquid handling, sample tracking, and data analysis. Through adding automation, particularly in the area of expert systems data review, the FBI Laboratory's Federal DNA Database Unit (FDDU) was able to process 408,000 DNA samples in a single year (October 2009 to September 2010) in order to eliminate their backlog (Richard Guerrieri, FBI FDDU Unit Chief, personal communication).

Liquid Sample Handling Robots

Robotic liquid-handling platforms for aiding sample preparation enables higher throughput, increases laboratory efficiency, and aids quality assurance efforts. Robotic systems can reduce hands-on time for DNA analysts enabling them to focus on other aspects of the process such as data interpretation. Potential sample contamination or mix-ups by human operators can be reduced because instruments will perform mundane tasks of sample transfers and deliver precise volumes of liquids more quickly and reliably.

There are a number of liquid handling tasks performed in DNA typing laboratories during the DNA extraction, PCR setup, and PCR amplification analysis steps. These liquid handling tasks are typically performed with manual pipettes by a DNA technician or analyst. Small volumes of liquids are repeatedly moved from one tube to another. These repetitive tasks can lead to mistakes as laboratory personnel become fatigued or careless.

By introducing automated liquid handling with robotics, the level of human error can be greatly reduced. Computers and robotics do the same task the same way time after time without tiring. The challenge though lies in setting up the automation and maintaining it (Hale 1999). Sample-handling automation is being used regularly with the high-volume sample processing of convicted offender samples for computer DNA databases (see Chapter 8).

As of early 2011, a variety of small-scale and large-scale liquid-handling robotic platforms are available with new ones in development. Some of the robotic systems in use by forensic DNA laboratories include: the Maxwell 16 (Promega Corporation), the EZ1 and QIAcube (QIAGEN), the Janus Automated Workstation (Perkin-Elmer), the Tecan Freedom EVO (Tecan) (Figure 17.4), and the Biomek 2000 and Biomek FX (Beckman Coulter). New sample



FIGURE 17.4 Tecan Freedom EVO 150 robotic workstation used for preparation of capillary electrophoresis sample plates. Photo courtesy of FBI Laboratory Federal DNA Database Unit.

preparation chemistries for improved DNA extraction will likely be incorporated into these and future robotic sample workstations.

Each robotic system has different capabilities and should be carefully assessed in order to meet the needs and goals of one's own laboratory environment (see Crouse & Conover 2003).

The Beckman Biomek 2000 robot has been used in conjunction with DNA IQ chemistry from the Promega Corporation to enable automated isolation and quantification of DNA samples (Tereba et al. 2003). Sexual assault (mixed stain) samples, cigarette butts, blood stains, buccal swabs, and various tissue samples were successfully extracted with the Biomek 2000 and the DNA IQ system without any evidence of contamination throughout extensive validation studies (Greenspoon et al. 2004). Robotic liquid handling for the steps of DNA extraction, quantification, PCR amplification setup, and preparation of sample plates for STR typing will likely become more prevalent in forensic laboratories particularly as the need for higher-volume work increases.

Laboratory Information Management Systems (LIMS)

Managing large amounts of data becomes a problem for many laboratories as they scale up their efforts. Computer databases are often developed to aid in tracking samples and results obtained. Sample tubes can be bar-coded and tracked through the analysis process. An example of efforts in this area is the Overlord System developed at the Forensic Science Service (FSS; Hopwood et al. 1997). The FSS Overlord program is a laboratory information management system (LIMS) and aids sample tracking as well as overall control of the different robotic stations. LIMS systems are rather expensive and are typically used only by laboratories with very high sample volumes.

Commercial LIMS systems, such as the Crime Fighter B.E.A.S.T. (computerized Bar-coded Evidence Analysis, Statistics, and Tracking LIMS) from Porter Lee Corporation (Schaumburg, IL), are being used in a growing number of forensic laboratories to provide electronic case files and automated sample tracking capabilities. A LIMS manufacturer typically sets up their software and customizes it to accommodate protocols and processes within each customer laboratory.

The Armed Forces DNA Identification Laboratory (AFDIL, Rockville, MD) has worked in conjunction with Future Technologies Inc. (Fairfax, VA) to develop LISA, which stands for Laboratory Information Systems Applications. LISA contains a number of sub-systems that permit case accessioning and the ability to electronically track the life cycle of each evidence and reference sample. There are additional modules such as MFIMS (Mass Fatality Incident Management System) and ASAP (AFDIL Statistical Application Program) that manage victim and family reference data as well as easing the tedious process of reporting results.

STaCS (Sample Tracking and Control System) is a system co-developed by forensic DNA scientists at the Royal Canadian Mounted Police (RCMP) and Anjura Technology Corporation (Ottawa, Ontario) that integrates robotic sample processing with custom LIMS software. STaCS monitors instrument performance and can provide a variety of operational information reports to help make the process of DNA typing more efficient. This system has been set up in several DNA databasing laboratories including the Florida Department of Law Enforcement (Tallahassee, FL) and the FBI Laboratory (Quantico, VA).

Fully integrated systems with robotic liquid handling are especially useful for DNA databasing of convicted offender samples, which are usually more uniform in nature (i.e., are all bloodstains or buccal swabs), single-source samples, and relatively concentrated in amount. In one laboratory, over 17,000 DNA samples were processed in a 20-month period using robotics and LIMS with an overall typing success rate of 99.99 % (Parson & Steinlechner 2001).

While automation is being developed and implemented to robotically process and track samples through the steps of DNA extraction, quantitation, PCR amplification, and sample setup prior to electrophoretic separation, separate computer programs commonly referred to as "expert systems" are being constructed to enable automatic interpretation of STR alleles from the resulting electropherograms.

EXPERT SYSTEMS FOR STR DATA INTERPRETATION

One of the most laborious efforts in the process of typing STRs is the data interpretation stage. For many high-throughput laboratories, data assessment and interpretation of STRs represents approximately 50 % or more of the resource requirement to deliver final results for samples. In many cases, more time is actually spent evaluating the STR profiles than preparing and processing the sample. In order to reduce this resource requirement, software has been designed and implemented to aid or replace the traditional manual assessment.

Two of the first expert systems used operationally include FSS-i^3 developed by the Forensic Science Service in England (and sold by Promega Corporation in the U.S.) and TrueAllele developed by Mark Perlin of Cybergenetics (Pittsburgh, PA). Applied Biosystems (Foster City, CA) has also developed GeneMapper*ID-X*, a program that includes quality flags on data collected with the ABI 310 or 3130*xl*, or 3500 Genetic Analyzers. Following validation of the software performance, DNA profiles from samples given a "green light" are generally considered fine and therefore further manual review is not required. As noted in Table 8.4, specific versions of these three expert systems have been approved by the National DNA Index System for use in uploading data to the U.S. national DNA database.

These expert systems are designed to translate the electropherogram signal into a genotype compatible with a database. As these expert systems are developed and implemented, bottlenecks will shift to other areas in the DNA typing process and thus permit development of expert systems that can solve ever more complex and diverse problems. Several years ago the National Institute of Justice established an NIJ-Expert System Testbed (NEST) project at Marshall University (Huntington, WV) that has evaluated the various expert system platforms for both single-source and mixture interpretation. For more information on the NEST project, see http://forensics.marshall.edu/NEST/default.html.

STR Expert System Suite (STRess) and FSS i-cubed Software

In 1998, the Forensic Science Service (FSS) developed a data interpretation program called STRess (STR Expert System Suite) to aid their STR profile processing. Interpretation guidelines drawn from approximately 100,000 samples processed by the FSS and used by experienced operators were incorporated into the programming of STRess (Gill et al. 1996, Dunbar et al. 1998). FSS genotyping guidelines require that all samples are genotyped by

two independent operators to ensure accuracy of DNA typing results followed by a third operator to review allele calls and confirm that they are concordant. The aim of STRess was to reduce the amount of manual effort needed to evaluate the STR data by replacing one of the genotype analysts. The FSS has estimated that incorporating the STRess program into routine analysis has resulted in a 10% to 20% time savings at the interpretation stage with improved standardization and quality of interpretation (Martin Bill, Forensic Science Service, personal communication).

The success of the original STRess program spawned the development of a number of systems that automate interpretation and interact as a suite configurable to any multiplex. The FSS-i^3 expert system has three modules: i-STRress for single-source sample review, i-STRream for interpreting two-person mixtures, and i-ntegrity for quality control data review (Bill & Knox 2005, Huston 2007).

Introduction of the suite of expert systems described above has resulted in a significant increase in efficiency and quality at the FSS with a large reduction in unit cost. Based largely on issues discovered during the development of the FSS-i^3 software, Martin Bill of the Forensic Science Service offers the following seven recommendations for expert system development:

1. **Integration** – Ensure that the information technology (IT) infrastructure, support, and storage issues are considered when designing and developing expert systems rather than concentrating solely on the interpretation aspects. Solutions that are selected without considering these IT issues may result in most of the financial benefits of the expert systems being lost in future IT expenditure.
2. **External influences** – Consider potential changes to the supply chain and ensure the system will still be able to perform as required. On occasion external influences may require a change to interpretation. Any expert system solution must be flexible enough to work around such changes without causing significant problems.
3. **Process design** – Implement a new process that encompasses the expert system; do not simply implement an expert system using existing protocols. Process re-engineering is invariably required to maximize benefits when implementing the expert system. If this is not considered, the benefits of the software may disappear and it can be difficult to undo the damage.
4. **Benefits measurement** – There is a trite statement "What gets measured gets done." Decide how the benefits of an expert system can be measured and ensure the measurement process takes place. Make sure the correct units of measurement are used. Remember you are trying to measure the actual benefits realized not just the potential of the software. Unit costs are invariably better indicators than timing exercises.
5. **People and culture** – Expect cultural issues until the scientists gain confidence in the expert system and include this aspect of the implementation in the project plan. Do not be surprised or disappointed if people need time to become accustomed to the idea of automated interpretation.
6. **Success rate** – This factor is often overlooked yet it is one of the most critical areas to consider. As a DNA unit moves from manual to automated interpretation, the success rate of the process will change. Expert systems will probably never have total concordance with manual interpretation because the computer is following a rigid set of rules. The change in success rate should be closely monitored during the initial phase of deployment.

7. **Target setting** – Set realistic targets for the project. Analysts do not spend all of their time analyzing data and therefore it is impossible to realize 100% analyst reduction irrespective of how good the expert system is. This aspect closely links with the process re-engineering. The same problem exists with projected error rates. Many laboratories refuse to acknowledge that an error rate exists. We should recognize that there are many opportunities for error to occur, some within and some outside the control of the DNA unit. It is better to openly acknowledge that error can occur as it is easier to look for solutions. *No expert system will ever be designed that has an error rate of zero* and therefore setting a target of zero is self-defeating. The real benefit of expert systems is that they behave predictably. It is this predictability and standardization that improves quality. As a starting point the objective should be to improve on the manual error rate (therefore making forward progress). When using expert systems, error rate and success rates are closely linked, one effectively determines the other. This level of control is extremely useful when attempting to optimize the output from a DNA unit.

TrueAllele

TrueAllele is a commercially available allele-calling program from Cybergenetics (Pittsburgh, PA) that uses quantitation and deconvolution algorithms to improve STR allele calls based upon quality measures (Pálsson et al. 1999, Perlin 2000). TrueAllele is written in Matlab and runs with Macintosh, Windows, or UNIX-based systems. The quality value assigned by TrueAllele ranges between 0.0 and 1.0 and reflects a peak's height, shape, and stutter pattern (Pálsson et al. 1999). The selection criteria used by the program are empirically derived through review of many STR profiles.

The first two forensic DNA laboratories to utilize TrueAllele as an expert system for STR analysis are the Forensic Science Service and the New York State Police in Albany, New York. In a comparison of allele calls from 2048 STR profiles between manual review with Genotyper and automated review with TrueAllele, only one significant difference was observed when the analyst using Genotyper interpreted a spike as a DNA peak at D8S1179 but TrueAllele correctly designated it as a spike (Kadash et al. 2004). Newer versions of TrueAllele are also capable of DNA mixture interpretation (Perlin et al. 2011).

GeneMapper*ID* and GeneMapper*ID-X*

Applied Biosystems (Foster City, CA) released a computer program in November 2003 named GeneMapper*ID* v3.1 (and later v3.2 and v3.2.1) that designates peaks in electropherograms and calls alleles through size comparisons to allelic ladders (Applied Biosystems 2003). GeneMapper*ID* still requires manual review of the data but process quality values (PQVs) provide confidence in allele calls and aid troubleshooting efforts. More recently, GeneMapper*ID-X* v1.1 and v1.2 software enables expert system processing of single-source samples and expert assistance support for DNA mixture interpretation.

Other DNA Interpretation Software Programs

The National Center for Biotechnology Information (NCBI) at the U.S. National Institutes of Health has developed a STR data interpretation software program called OSIRIS, which stands

for Open Source Independent Review and Interpretation System. It is freely available for download in Mac or Windows formats at http://www.ncbi.nlm.nih.gov/projects/SNP/osiris/.

GenoProof 2 and GenoProof Mixture are STR data interpretation programs developed by Qualitype AG (Dresden, Germany) that enable information processing from raw data to biostatistical calculations. The GenoProof Mixture program was especially designed to handle complex DNA mixtures. For more information, see http://www.qualitype.de.

Soft Genetics, LLC (State College, PA) has developed STR typing software for human identity applications that can perform mixture analysis and database searches as well as paternity and kinship analysis (Holland & Parson 2011). More information is available at http://www.softgenetics.com/GeneMarkerHID.html.

SUMMARY AND FINAL THOUGHTS

The adoption of new technology by the forensic DNA community takes time (e.g., several years) for multiple reasons. First and foremost, methods need to be carefully validated to ensure that results with a new technology are accurate and reproducible (see Chapter 7). Second, methods should yield comparable results to current technologies so that genotype information can be compared over time. The development of large DNA databases make it necessary to have a constant currency so that convicted offender samples have been analyzed with the same DNA markers as crime scene samples (see Chapter 8).

A new set of markers or a new form of sample analysis, unless it gives an equivalent result to current technology, must have clear advantages and be very inexpensive to overcome legacy data in large DNA databases. Now that millions of DNA profiles are present

TABLE 17.2 Benefits and Limitations of Several Technologies for DNA Testing.

Technology	Benefits	Limitations
Rapid DNA analysis	Potentially portable system; faster results that could enable new applications of DNA testing	Some compromises in DNA result quality may be required
Mass spectrometry base composition	Enables measurement of both length and nucleotide variability with compatibility to existing STR databases; less expensive than sequencing for mtDNA	Higher instrument cost and more difficult to maintain than a CE system; unable to determine the location of a sequence change; cannot multiplex to a high-level (would likely require more DNA)
Pyrosequencing	Enables rapid sequence-based analysis of STRs or other PCR products	Cannot multiplex (would likely require more DNA); cannot cope with complex and long repeats
Next-generation DNA sequencing	Enables recovery of sequence information from highly degraded DNA (e.g., ancient DNA samples)	Significant increase in cost compared to current methods; short read lengths limit ability to capture STR repeat variation
Automation	Improved efficiency and speed in processing samples (particularly with expert systems data interpretation)	Initial cost for equipment and software

in national DNA databases it is highly unlikely that the field will abandon the current STR loci in the near-to-medium future. Yet as noted throughout this book, the continued progress being made in biotechnology around the world has led and will continue to lead to improved methods for DNA typing. Improvements in expert systems and data interpretation computer programs—particularly with the ability to handle DNA mixtures and low-level samples—are an important area for the future.

A brief review of the benefits and limitations of each technology discussed in this chapter is contained in Table 17.2. Each potential new method has advantages and disadvantages. Faster DNA analysis, higher levels of sensitivity, and stronger powers of discrimination are possible as new technology is embraced and adopted following validation by the forensic DNA typing community. Automation is also a key to the future—particularly in terms of data interpretation.

READING LIST AND INTERNET RESOURCES

Rapid DNA Analysis with Microchip CE Systems

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